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Section A

Q1
→ Hardware - the tangible physical parts of a computer.
Examples: CPU, SSD.
Software - the intangible programs and data that tell the hardware what to do.
Examples: OS, Spotify.

Q2
→ OS is a layer between user programs and hardware.
1. Manages resources
2. Provides abstraction
3. Controls hardware via drivers.

Q3
→ Machine code is a stream of binary instructions (0s and 1s) that a CPU can execute directly.
→ High level languages (Python, C++) must be translated into machine code because a processor can only understand its own ins. set architecture.

Q4

(faster) Compiler	(slower) Interpreter
converts entire source file to machine code before exe.	Translates and executes line by line at runtime.
Stand alone binary	No separate binary.

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RAM	SSD.
volatile	non-volatile
nanoseconds access	milliseconds or micro seconds access
working memory for active programs and data.	long term storage.

Q5 $(110101)_2 \rightarrow (\ )_{10}$.

$1(2^5)+1(2^4)+0(2^3)+1(2^2)+0(2^1)+1(2^0)$
$=(53)_{10}$

Q6 ADC

→ Sampling - Measure the analog signals amplitude at uniform time intervals

→ Quantization - Round each samples amplitude to the nearest level representable with a fixed number of bits.

→ Encoding - Store each quantized value as a binary word. The resulting bit stream is the digital version of the original analog signal.

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Q7

→ Significance : Limits the SNR and fidelity of digital audio, images, etc.

→ Minimization : Use more bits per sample (finer steps), apply dithering or use noise shaping techniques.

Q8

→ Readable, concise syntax - speeds up prototyping and collaboration.

→ Rich ecosystem - libraries like Numpy, pandas, scikitlearn, Tensor flow, PyTorch.

→ Large community & resources - abundant tutorials, forums, and pre trained models.

Q9

→ .py → plain text python script

.ipynb → Jupyter notebook file, Json format containing code cells, output, and markdown, executed interactively in a browser.

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Q10
→ Interactive mode → type commands at a prompt (>>>) or in a Jupyter cell and see immediate results. Best for quick testing, exploration and data analysis.
→ Script mode → write code in .py and run it as a whole. Best for larger programs, automation and version control matter.

Section B

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12 C
13 C
14 A
15 C
16 A
17 B
18 C
19 C
20 C